HAWK Fakultät Ingenieurwissenschaften und Gesundheit Göttingen

POLITECNICO MILANO 1863

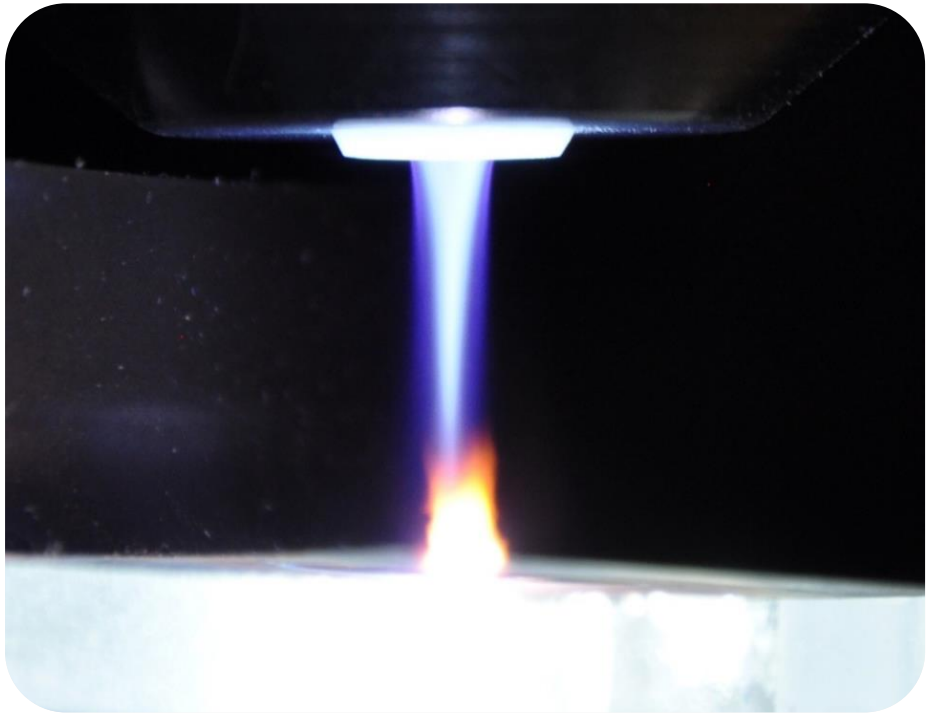
Exercises and Worked Examples + Solutions

ADVANCED TOPICS IN ENGINEERING PHYSICS

'Plasma and its applications'
Chapter 4

Prof. Dr. Christoph Gerhard, 2024

4 Plasmas in photonics



Plasma shielding and focussing

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- **Exercise 4.1**
- Determine the index of refraction of a plasma with an electron density of $10^{15}/m^3$ for incident CO_2-laser irradiation.

Plasma shielding and focussing

! ! !

- **Exercise 4.1 – Solution**
- The plasma frequency amounts to

$$\omega_p = \sqrt{\frac{(1{,}60218 \cdot 10^{-19} As)^2 \cdot \frac{10^{15}}{m^3}}{9{,}10938 \cdot 10^{-31} kg \cdot 8{,}85419 \cdot 10^{-12} \frac{As}{Vm}}} = 1.783.990.256 s^{-1} \approx 1{,}78 GHz$$

- The (angular) frequency of CO_2-laser irradiation (λ = 10.6 µm) is

$$\omega_{laser} = 2 \cdot \pi \cdot f = \frac{c}{\lambda} \approx 177.7 THz.$$

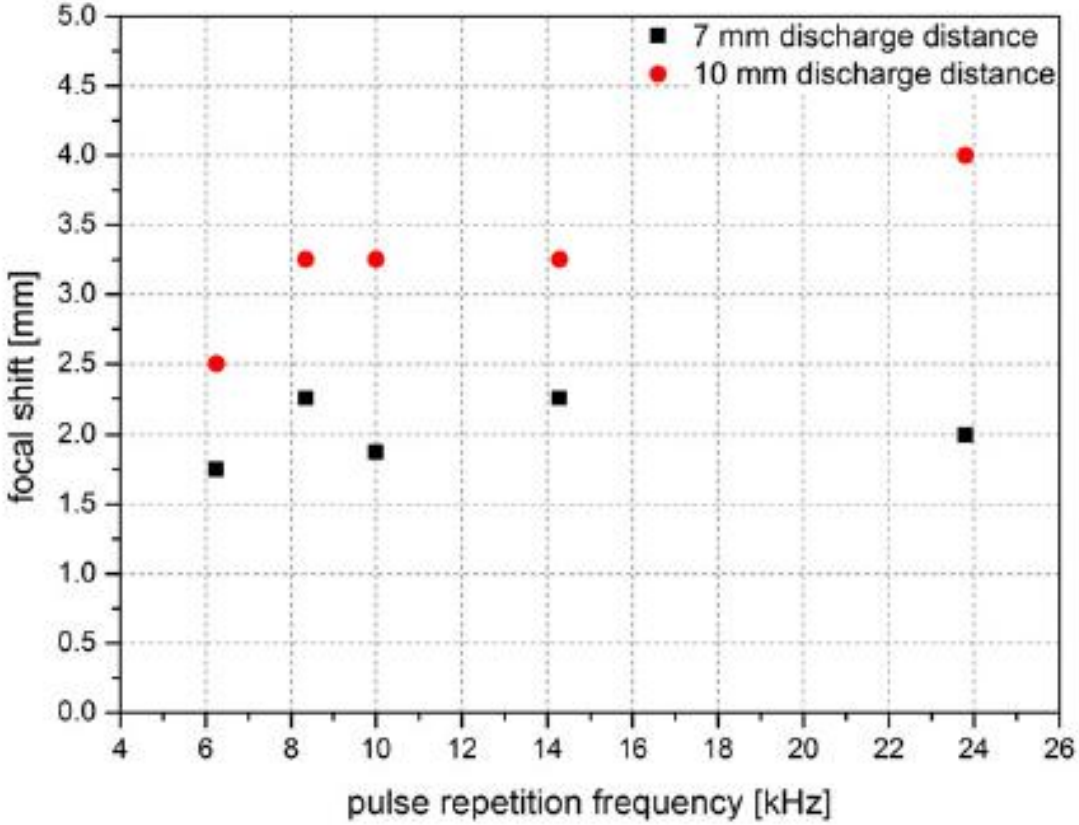
- The index of refraction of the plasma for this laser irradiation is thus

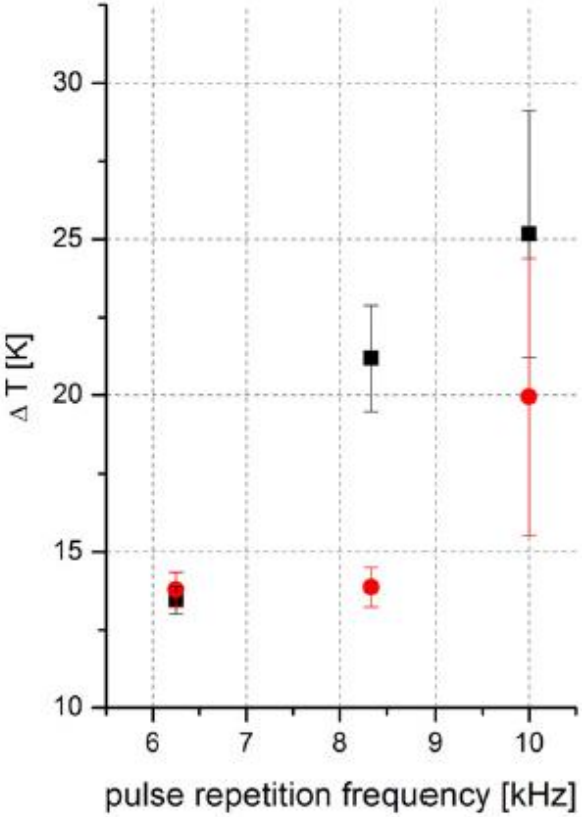
$$n = \sqrt{1 - \frac{\omega_p^2}{\omega_{laser}^2}} = 0.99.$$

- → **The index of refraction of a plasma can – theoretically – be smaller than 1, i.e., the index of refraction of vacuum!**

Plasma shielding and focussing

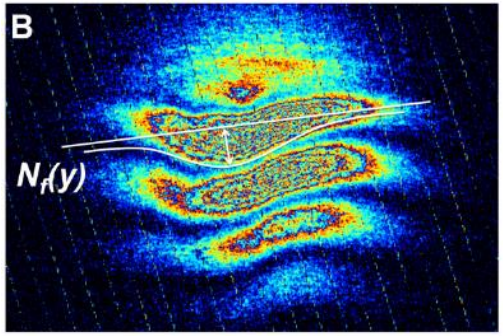
- **Worked example 4.1**
- What is the reason for the focal shift shown below? Identify the underlying mechanism/physical effect…
  Remember: the laser beam propagates coaxially to the plasma beam ('common path')



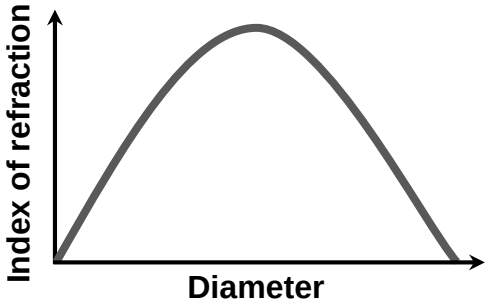


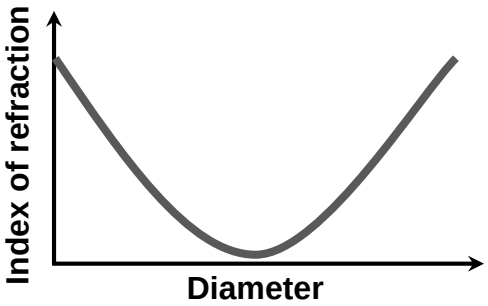
Plasma shielding and focussing

- **Worked example 4.1 – Answer**
- The plasma features a – more or less – Gaussian temperature distribution (see interferogram) and thus a Gaussian distribution in index of refraction (see Edlén equation)



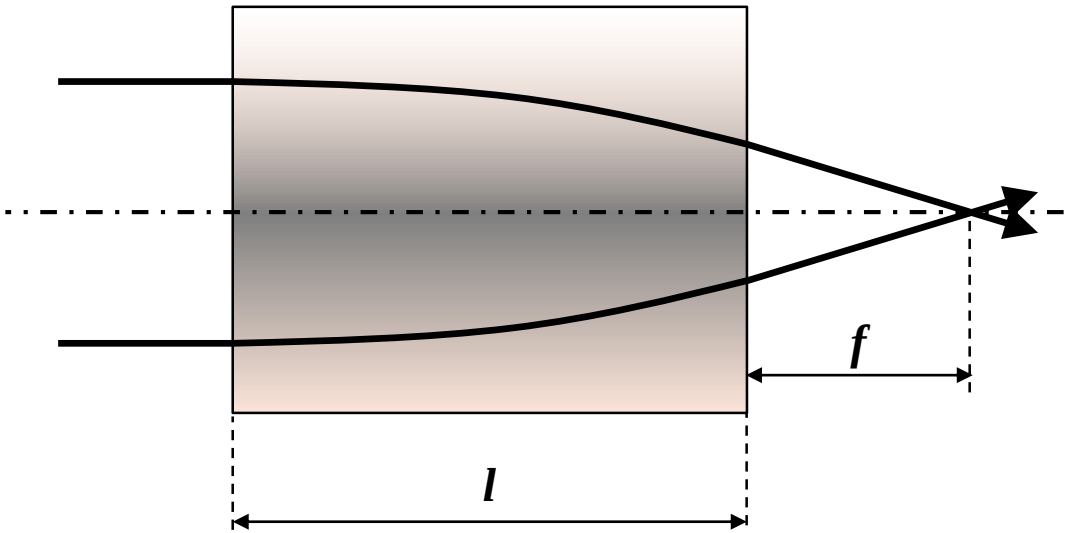
  - → The plasma temperature decreases with increasing distance form the centre axis
  - → The index of refraction of the plasma increases with increasing distance form the centre axis
  - → The plasma represents a so-called gradient index (GRIN) material, featuring a gradient index constant *g* (≈ slope of change in index of refraction over diameter):





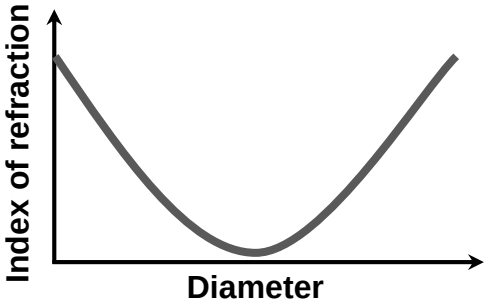
Plasma shielding and focussing

- **Worked example 4.1 – Answer**
- The plasma acts like a gradient index lens (GRIN lens) and thus features an inherent focal length:



$$f = \frac{1}{n_0 \cdot g \cdot \sin(g \cdot l)}$$

- Incoming light is thus focussed/defocussed, depending on the distribution in index of refraction
- Here: defocussing (-f), since:



Plasma shielding and focussing

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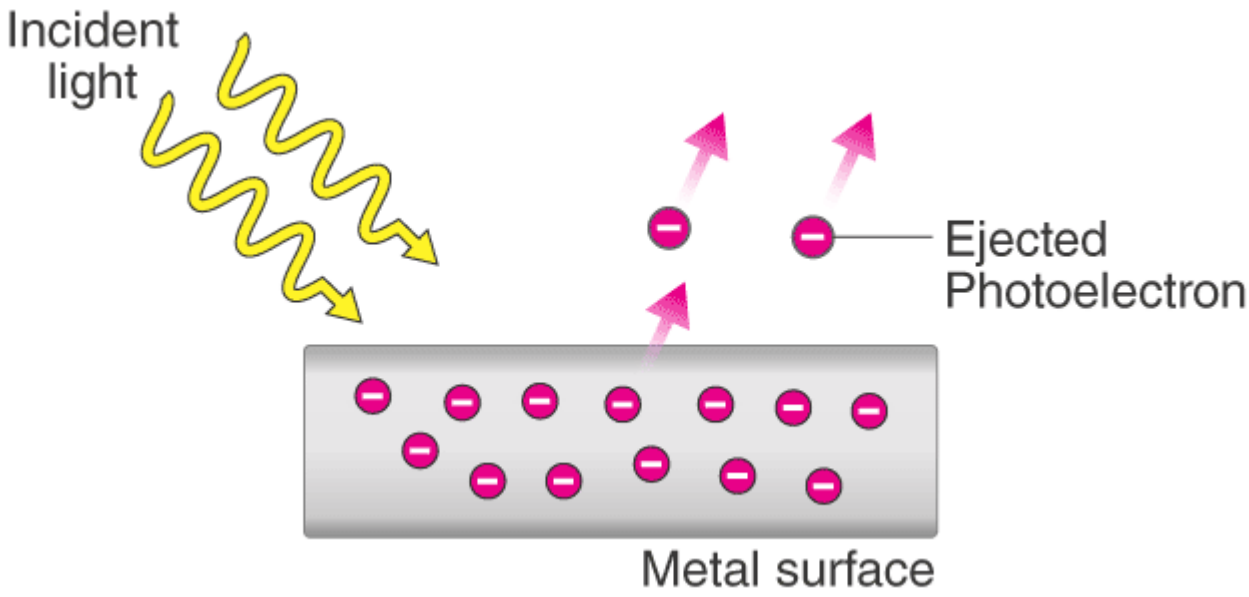
- **Exercise 4.2**
- Determine the energy of an electron which is released from a vanadium surface via the photoelectric effect induced by a V-UV photon emitted by the 1s-2p transition of sevenfold ionised nitrogen.



Plasma shielding and focussing

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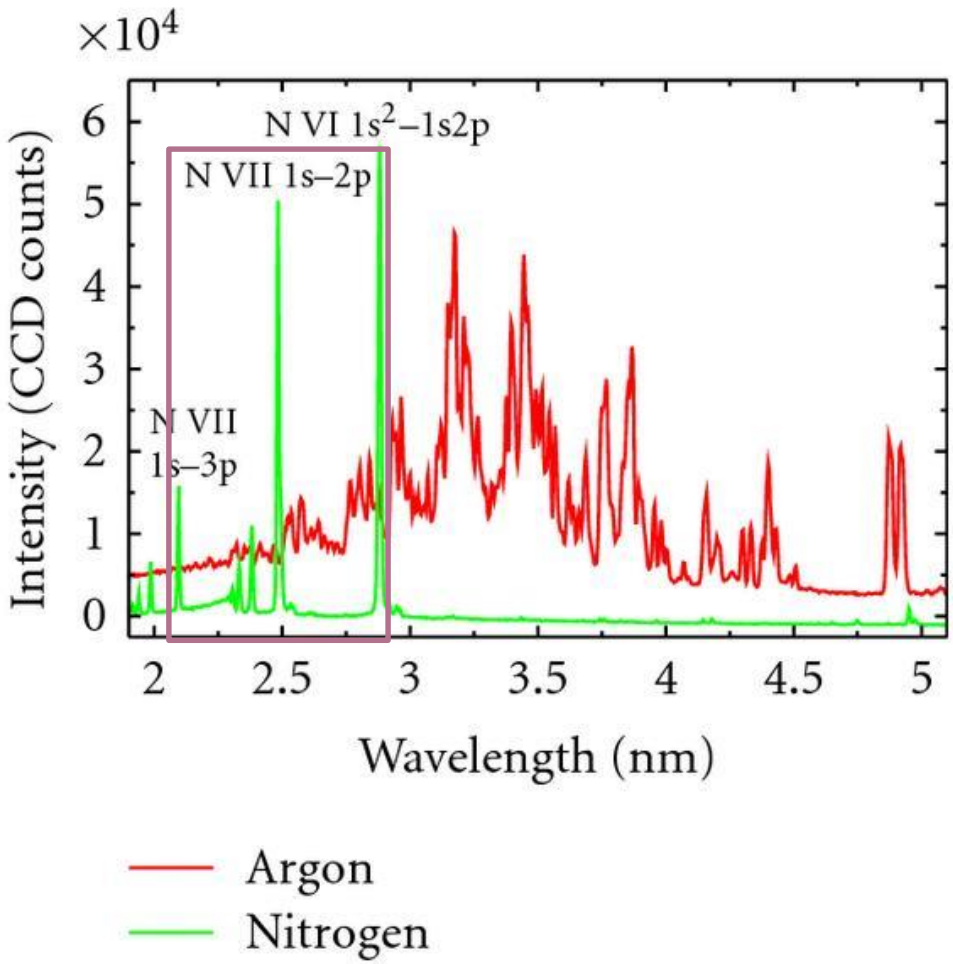
- **Exercise 4.2 – Solution**
- The wavelength of the photon is about 2.5 nm, see graph:



Plasma shielding and focussing

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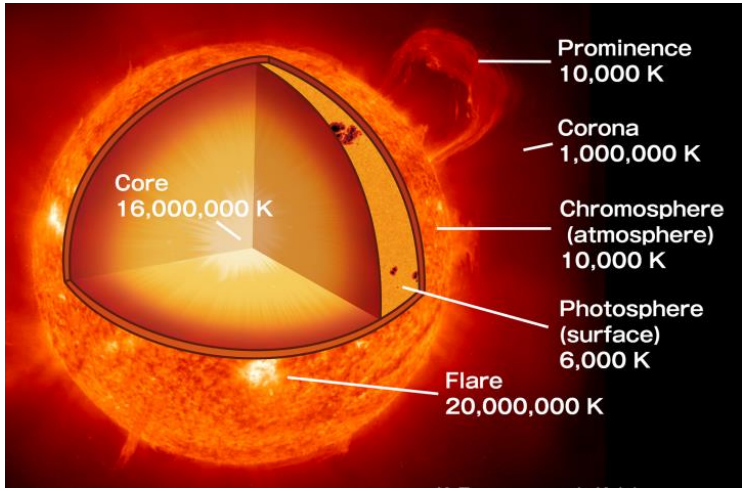
- **Exercise 4.2 – Solution**
- The photon energy is thus 495.94 eV (or 79.46 Attojoule, respectively)!
- The work function of vanadium amounts to 4.3 eV [1].
- The energy of the electron released by the V-UV photon is thus

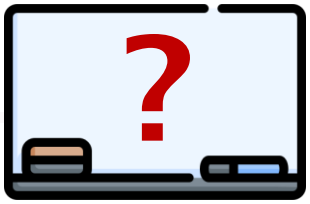
$$E_{e-} = E_{kin} = 495.94eV - 4.3eV = 491.64eV.$$

- Note: Assuming that 1 eV = 11604K, the electron temperature is 5.7MK, i.e., approx. a third of the core temperature of Sun.

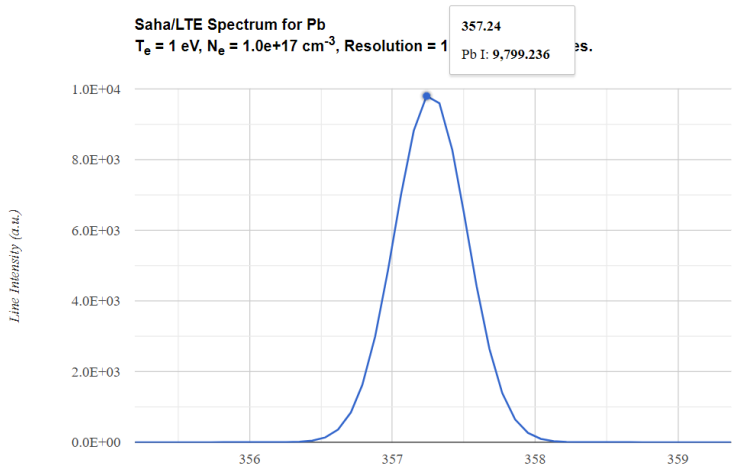


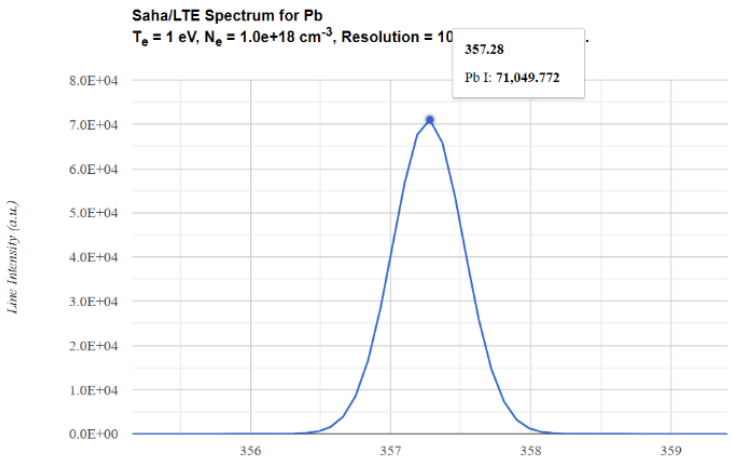
[1] CRC Handbook of Chemistry and Physics version 2008, p. 12–114, published on Wikipedia, see: https://en.wikipedia.org/wiki/Work_function

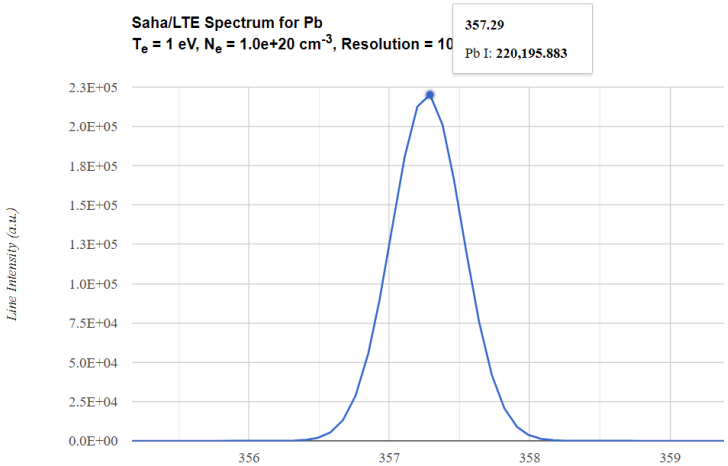
Evaluation of laser-induced plasmas

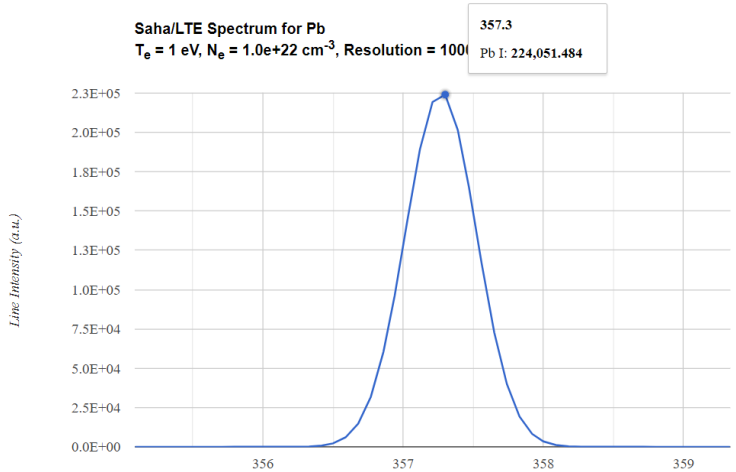


- **Worked example 4.2**
- Which effect can be observed below…?



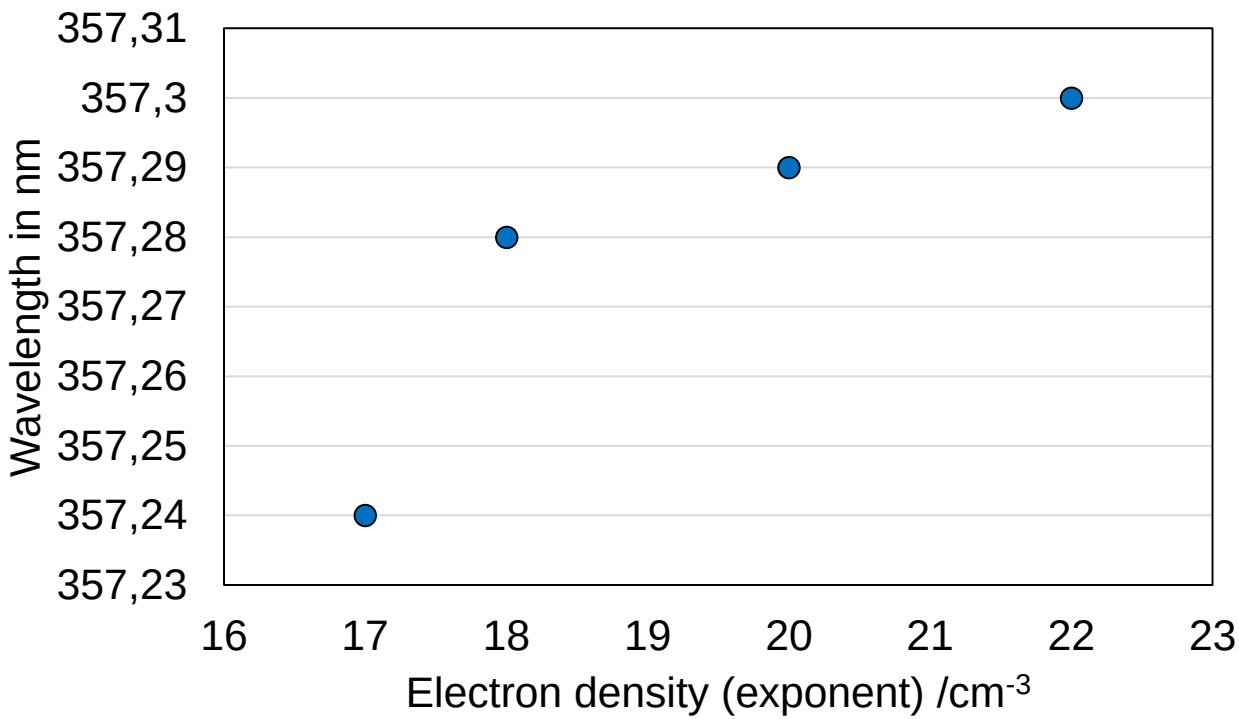






Evaluation of laser-induced plasmas

- **Worked example 4.2 – Answer**
- We observe that for increasing electron density, the central wavelength of the (simulated) signal increases:



- This effect is referred to as the Stark shift.

Evaluation of laser-induced plasmas

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- **Exercise 4.3**

a) Determine the frequency bandwidth Δf of light emitted by the sodium Na I transition at a wavelength of 588.995095 nm.

b) Determine the lifetime of the excited state of this transition assuming spontaneous emission (as occurring in LIBS).

Evaluation of laser-induced plasmas

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- **Exercise 4.3**

a) The Einstein coefficient of this transition amounts to 6.16 x 10^7/s:

Ion	Observed Wavelength Air (nm)	Unc. (nm)	Ritz Wavelength Air (nm)	Unc. (nm)	Rel. Int. (?)	A_{ki} (s^{-1})	Acc.	E_i (cm^{-1})		E_k (cm^{-1})	Lower Level Conf., Term, J	Upper Level Conf., Term, J
Na I	588.995095	0.000003	588.995094	0.000003	80000	6.16e+07	AA	0.00000	-	16 973.36619	$2p^6 3s$ ^2S $^1/_2$	$2p^6 3p$ $^2P°$ $^3/_2$
Na VII			589	5		3.09e+03	D	1 415 630	-	1 432 600?	$2p^2(^1D)3d$ ^2D $^3/_2$	$2s2p(^3P°)4s$ $^2P°$ $^3/_2$
Na VII			589	5		2.79e+04	D+	1 415 630	-	1 432 600?	$2p^2(^1D)3d$ ^2D $^5/_2$	$2s2p(^3P°)4s$ $^2P°$ $^3/_2$
Na I	589.592424	0.000003	589.592424	0.000003	40000	6.14e+07	AA	0.00000	-	16 956.17025	$2p^6 3s$ ^2S $^1/_2$	$2p^6 3p$ $^2P°$ $^1/_2$

- The frequency bandwidth Δf is thus

$$\Delta f = \frac{1}{\tau} \cdot \frac{1}{2 \cdot \pi} = A_{ki} \cdot \frac{1}{2 \cdot \pi} = 6.16 \cdot \frac{10^7}{s} \cdot \frac{1}{2 \cdot \pi} \approx 9.8 MHz$$

b)

The lifetime of the excited state τ is ‘coded’ in the Einstein coefficient. Solving $\Delta f = \frac{1}{\tau} \cdot \frac{1}{2 \cdot \pi}$ for τ gives $\tau = \frac{1}{\Delta f} \cdot \frac{1}{2 \cdot \pi}$.

The residence time of the electron in the excited level is thus $\tau = \frac{1}{9.8 MHz} \cdot \frac{1}{2 \cdot \pi} = 1.624 \cdot 10^{-8} s \approx 16 ns$.